

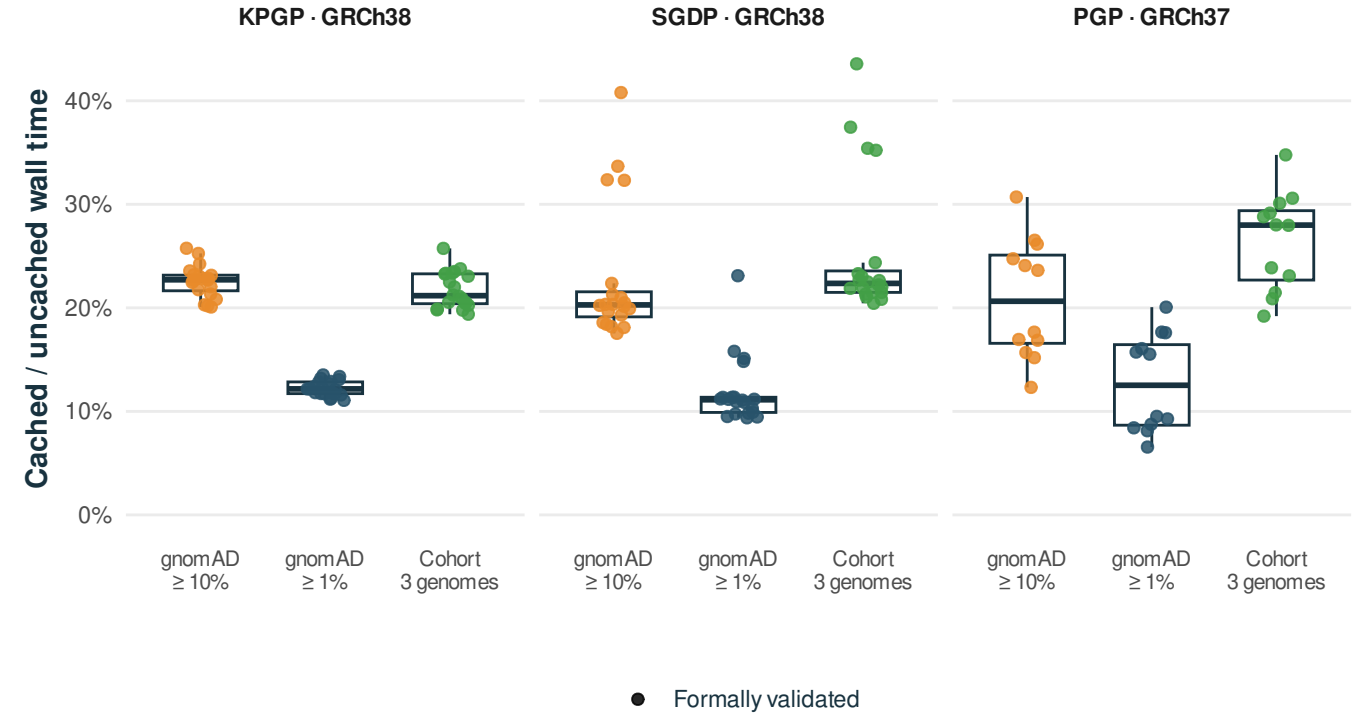
# Real-world WGS: bundled versus cohort-derived caches

KPGP, SGDP and PGP · no documented gnomAD project overlap · one run per sample

A

## Runtime remaining by cache strategy

No documented gnomAD project overlap; held out from cohort-cache construction



B

## Observed behavior follows the miss fraction

Dashed line: zero-overhead rule; vertical excess is lookup and preprocessing



C

## A custom cache repays its one-time build cost

Effective speedup includes measured three-genome cache construction

